

5.5. TWO-DIMENSIONAL RANDOM VARIABLES

So far we have defined only one random variable on a sample space. It is also possible to define more than one random variable on the same sample space. For example, we may be interested in recording the height and weight of every person in a certain educational institution. To describe such experiments mathematically we introduce the study of two random variables.

Definition. Let X and Y be two random variables defined on the same sample space S , then the function (X, Y) that assigns a point in $R^2 (= R \times R)$, is called a two-dimensional random variable.

Let (X, Y) be a two-dimensional random variable, defined on the sample space S and $\omega \in S$. The value of (X, Y) at ω is given by the pair of real numbers $\{X(\omega), Y(\omega)\}$. The notation $\{X \leq a, Y \leq b\}$ denotes the event of all elements $\omega \in S$, such that $X(\omega) \leq a$ and $Y(\omega) \leq b$. The probability of the event $\{X \leq a, Y \leq b\}$ will be denoted by $P(X \leq a, Y \leq b)$.

Let $A = \{a < X \leq b\}$, $B = \{c < Y \leq d\}$ be two events. Then the event

$$\{a < X \leq b, c < Y \leq d\} = \{a < X \leq b\} \cap \{c < Y \leq d\} = A \cap B$$

$$\therefore P(a < X \leq b, c < Y \leq d) = P(A \cap B)$$

Remarks:

1. A two-dimensional random variable is said to be discrete if it takes at most a countable number of points in R^2 .

2. Two random variables X and Y are said to be jointly distributed if they are defined on the same probability space. The sample points consist of 2-tuples. If the joint probability function is denoted by $P_{XY}(x, y)$ then the probability of a certain event E is given by :

$$P_{XY}(x, y) = P[(X, Y) \in E]$$

... (5-13)

5.5.1. Two-dimensional or Joint Probability Mass Function

Let X and Y be random variables on a sample space S with respective image sets

$X(S) = \{x_1, x_2, \dots, x_n\}$ and $Y(S) = \{y_1, y_2, \dots, y_m\}$. We make the product set

$$X(S) \times Y(S) = \{x_1, x_2, \dots, x_n\} \times \{y_1, y_2, \dots, y_m\}$$

into a probability space by defining the probability of the ordered pair (x_i, y_j) to be $P(X = x_i, Y = y_j)$ which we write $p(x_i, y_j)$. The function p on $X(S) \times Y(S)$ defined by :

$P_{ij} = P(X = x_i \cap Y = y_j) = p(x_i, y_j)$ is called the *joint probability function* of X and Y and is usually represented in the form of the following table :

$Y \backslash X$	y_1	y_2	y_3	$\dots$	y_j	$\dots$	y_m	Total
x_1	p_{11}	p_{12}	p_{13}	$\dots$	p_{1j}	$\dots$	p_{1m}	$p_{1.}$
x_2	p_{21}	p_{22}	p_{23}	$\dots$	p_{2j}	$\dots$	p_{2m}	$p_{2.}$
x_3	p_{31}	p_{32}	p_{33}	$\dots$	p_{3j}	$\dots$	p_{3m}	$p_{3.}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_i	p_{i1}	p_{i2}	p_{i3}	$\dots$	p_{ij}	$\dots$	p_{im}	$p_{i.}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_n	p_{n1}	p_{n2}	p_{n3}	$\dots$	p_{nj}	$\dots$	p_{nm}	$p_{n.}$
Total	$p_{.1}$	$p_{.2}$	$p_{.3}$	$\dots$	$p_{.j}$	$\dots$	$p_{.m}$	1

Definition. If (X, Y) is a two-dimensional discrete random variable, then the joint discrete function of X, Y , also called the *joint probability mass function* of X, Y , denoted by $p_{X, Y}$ is defined as :

$$p_{XY}(x_i, y_j) = P(X = x_i, Y = y_j) \text{ for a value } (x_i, y_j) \text{ of } (X, Y) \quad \dots (5-14)$$

and $p_{XY}(x_i, y_j) = 0$, otherwise.

Remark. It may be noted that $\sum \sum p_{XY}(x_i, y_j) = 1$, where the summation is taken over all possible values of (X, Y) .

Marginal Probability Function. Let (X, Y) be a discrete two-dimensional r.v. which takes up countable number of values (x_i, y_j) . Then the probability distribution of X is determined as follows :

$$p_X(x_i) = P(X = x_i) \\ = P(X = x_i \cap Y = y_1) + P(X = x_i \cap Y = y_2) + \dots + P(X = x_i \cap Y = y_m)$$

$$= p_{i1} + p_{i2} + \dots + p_{ij} + \dots + p_{im} = \sum_{j=1}^m p_{ij} = \sum_{j=1}^m p(x_i, y_j) = p_{i.} \quad \dots (5-14 a)$$

and is known as *marginal probability mass function or discrete marginal density function of X*.

Also,
$$\sum_{i=1}^n p_{i.} = p_{1.} + p_{2.} + \dots + p_{n.} = \sum_{i=1}^n \sum_{j=1}^m p_{ij} = 1$$

Similarly, we can prove that

$$p_Y(y_j) = P(Y = y_j) = \sum_{i=1}^n p_{ij} = \sum_{i=1}^n p(x_i, y_j) = p_{.j} \quad \dots (5-14 b)$$

which is the *marginal probability mass function of Y*.

Conditional Probability Function

Definition. Let (X, Y) be a discrete two-dimensional random variable. Then the conditional discrete density function or the conditional probability mass function of X , given $Y = y$, denoted by $p_{X|Y}(x|y)$, is defined as :

$$p_{X|Y}(x|y) = \frac{P(X = x, Y = y)}{P(Y = y)}, \text{ provided } P(Y = y) \neq 0 \quad \dots (5-14 c)$$

Since for a fixed y ,

$$\sum_i \frac{P(X = x_i, Y = y)}{P(Y = y)} = \frac{1}{P(Y = y)} \sum_i P(X = x_i, Y = y) = \frac{1}{P(Y = y)} P(Y = y) = 1,$$

it follows that the conditional discrete density function $f_{X|Y}(x|y)$ is a discrete density function, when considered as a function of the values of X .

The conditional probability mass function $p_{Y|X}(y|x)$ is similarly defined, i.e.,

$$p_{Y|X}(y|x) = \frac{P(X = x, Y = y)}{P(X = x)}$$

A necessary and sufficient condition for the discrete random variables X and Y to be independent is :

$$P(X = x_i, Y = y_j) = P(X = x_i) P(Y = y_j) \text{ for all values } (x_i, y_j) \text{ of } (X, Y) \quad \dots (5-14 d)$$

5.5.2. Two-Dimensional Distribution Function

Definition. The distribution function of the two-dimensional random variable (X, Y) is a real valued function F defined for all real x and y by the relation :

$$F_{XY}(x, y) = P(X \leq x, Y \leq y). \quad \dots (5-15)$$

Properties of Joint Distribution Function

- (i) For the real numbers a_1, b_1, a_2 and b_2 ; ($a_1 < b_1$; $a_2 < b_2$),

$$P(a_1 < X \leq b_1, a_2 < Y \leq b_2) = F_{XY}(b_1, b_2) + F_{XY}(a_1, a_2) - F_{XY}(a_1, b_2) - F_{XY}(b_1, a_2)$$

[For proof see Example 5-49]

(ii) From the adjoining diagram for $a_1 < b_1, a_2 < b_2$; we have

$$P(X \leq a_1, Y \leq a_2) + P(a_1 \leq X \leq b_1, Y \leq a_2)$$

$$= P(X \leq b_1, Y \leq a_2)$$

$$\Rightarrow P(X \leq b_1, Y \leq a_2) - P(X \leq a_1, Y \leq a_2)$$

$$= P[a_1 \leq X \leq b_1, Y \leq a_2] \geq 0$$

$$\Rightarrow F(b_1, a_2) - F(a_1, a_2) \geq 0$$

$$F(b_1, a_2) \geq F(a_1, a_2)$$

... (*)

Similarly it follows that :

$$F(a_1, b_2) - F(a_1, a_2) = P(X \leq a_1, a_2 \leq Y \leq b_2) \geq 0$$

$$F(a_1, b_2) \geq F(a_1, a_2)$$

... (**)

(*) and (**) show that $F(x, y)$ is monotonic non-decreasing function in both variables x and y .

$$F(-\infty, y) = 0 = F(x, -\infty), \quad F(+\infty, +\infty) = 1.$$

3. If the density function $f(x, y)$ is continuous at (x, y) ,

$$\frac{\partial^2 F}{\partial x \partial y} = f(x, y).$$

5.3. Marginal Distribution Functions

From the knowledge of joint distribution function $F_{XY}(x, y)$, it is possible to obtain the individual distribution functions, $F_X(x)$ and $F_Y(y)$ which are termed as marginal distribution function of X and Y , respectively, with respect to the joint distribution function $F_{XY}(x, y)$.

$$F_X(x) = P(X \leq x) = P(X \leq x, Y < \infty) = \lim_{y \rightarrow \infty} F_{XY}(x, y) = F_{XY}(x, \infty) \quad \dots (5.16)$$

$$\text{Similarly, } F_Y(y) = P(Y \leq y) = P(X < \infty, Y \leq y) = \lim_{x \rightarrow \infty} F_{XY}(x, y) = F_{XY}(\infty, y) \quad \dots 5.16(a)$$

$F_X(x)$ is termed as the marginal distribution function of X corresponding to the joint distribution function $F_{XY}(x, y)$ and similarly $F_Y(y)$ is called marginal distribution function of the random variable Y corresponding to the joint distribution function $F_{XY}(x, y)$.

In the case of jointly discrete random variables, the marginal distribution functions are given

$$F_X(x) = \sum_y P(X \leq x, Y = y), \quad \text{and} \quad F_Y(y) = \sum_x P(X = x, Y \leq y)$$

Similarly, in the case of jointly continuous random variables, the marginal distribution functions are given as :

$$F_X(x) = \int_{-\infty}^x \left\{ \int_{-\infty}^{\infty} f_{XY}(x, y) dy \right\} dx, \quad F_Y(y) = \int_{-\infty}^y \left\{ \int_{-\infty}^{\infty} f_{XY}(x, y) dx \right\} dy$$

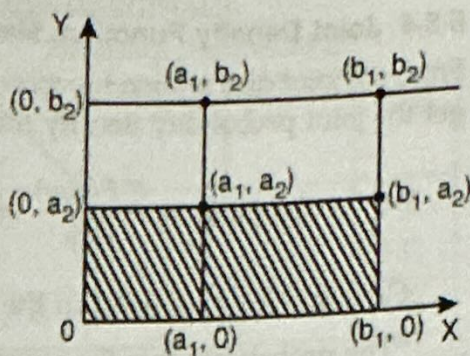


Fig. 5.9a

(ii) From the adjoining diagram for $a_1 < b_1, a_2 < b_2$; we have

$$\begin{aligned} P(X \leq a_1, Y \leq a_2) + P(a_1 \leq X \leq b_1, Y \leq a_2) \\ = P(X \leq b_1, Y \leq a_2) \\ \Rightarrow P(X \leq b_1, Y \leq a_2) - P(X \leq a_1, Y \leq a_2) \\ = P[a_1 \leq X \leq b_1, Y \leq a_2] \geq 0 \end{aligned}$$

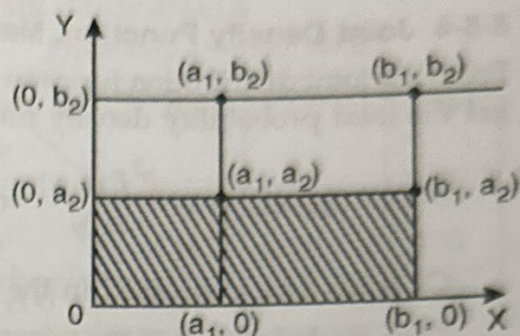


Fig. 5.9a

$$\Rightarrow F(b_1, a_2) - F(a_1, a_2) \geq 0$$

$$\Rightarrow F(b_1, a_2) \geq F(a_1, a_2) \quad \dots (*)$$

Similarly it follows that :

$$F(a_1, b_2) - F(a_1, a_2) = P(X \leq a_1, a_2 \leq Y \leq b_2) \geq 0$$

$$\Rightarrow F(a_1, b_2) \geq F(a_1, a_2) \quad \dots (**)$$

(*) and (**) show that $F(x, y)$ is monotonic non-decreasing function in both variables x and y .

$$2. \quad F(-\infty, y) = 0 = F(x, -\infty), \quad F(+\infty, +\infty) = 1.$$

$$3. \text{ If the density function } f(x, y) \text{ is continuous at } (x, y), \quad \frac{\partial^2 F}{\partial x \partial y} = f(x, y).$$

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From the knowledge of joint distribution function $F_{XY}(x, y)$, it is possible to obtain the individual distribution functions, $F_X(x)$ and $F_Y(y)$ which are termed as marginal distribution function of X and Y , respectively, with respect to the joint distribution function $F_{XY}(x, y)$.

$$F_X(x) = P(X \leq x) = P(X \leq x, Y < \infty) = \lim_{y \rightarrow \infty} F_{XY}(x, y) = F_{XY}(x, \infty) \quad \dots (5.16)$$

$$\text{Similarly, } F_Y(y) = P(Y \leq y) = P(X < \infty, Y \leq y) = \lim_{x \rightarrow \infty} F_{XY}(x, y) = F_{XY}(\infty, y) \quad \dots 5.16(a)$$

$F_X(x)$ is termed as the marginal distribution function of X corresponding to the joint distribution function $F_{XY}(x, y)$ and similarly $F_Y(y)$ is called marginal distribution function of the random variable Y corresponding to the joint distribution function $F_{XY}(x, y)$.

In the case of jointly discrete random variables, the marginal distribution functions are given as :

$$F_X(x) = \sum_y P(X \leq x, Y = y), \quad \text{and} \quad F_Y(y) = \sum_x P(X = x, Y \leq y)$$

Similarly, in the case of jointly continuous random variables, the marginal distribution functions are given as :

$$F_X(x) = \int_{-\infty}^x \left\{ \int_{-\infty}^{\infty} f_{XY}(x, y) dy \right\} dx, \quad F_Y(y) = \int_{-\infty}^y \left\{ \int_{-\infty}^{\infty} f_{XY}(x, y) dx \right\} dy$$

(ii) From the adjoining diagram for $a_1 < b_1, a_2 < b_2$;
we have

$$\begin{aligned} P(X \leq a_1, Y \leq a_2) + P(a_1 \leq X \leq b_1, Y \leq a_2) \\ = P(X \leq b_1, Y \leq a_2) \\ \Rightarrow P(X \leq b_1, Y \leq a_2) - P(X \leq a_1, Y \leq a_2) \\ = P[a_1 \leq X \leq b_1, Y \leq a_2] \geq 0 \end{aligned}$$

$$\Rightarrow F(b_1, a_2) - F(a_1, a_2) \geq 0$$

$$\Rightarrow F(b_1, a_2) \geq F(a_1, a_2) \quad \dots (*)$$

Similarly it follows that :

$$F(a_1, b_2) - F(a_1, a_2) = P(X \leq a_1, a_2 \leq Y \leq b_2) \geq 0$$

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(*) and (**) show that $F(x, y)$ is monotonic non-decreasing function in both variables x and y .

$$2. \quad F(-\infty, y) = 0 = F(x, -\infty), \quad F(+\infty, +\infty) = 1.$$

$$3. \quad \text{If the density function } f(x, y) \text{ is continuous at } (x, y), \quad \frac{\partial^2 F}{\partial x \partial y} = f(x, y).$$

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In the case of jointly discrete random variables, the marginal distribution functions are given as :

$$F_X(x) = \sum_y P(X \leq x, Y = y), \quad \text{and} \quad F_Y(y) = \sum_x P(X = x, Y \leq y)$$

Similarly, in the case of jointly continuous random variables, the marginal distribution functions are given as :

$$F_X(x) = \int_{-\infty}^x \left\{ \int_{-\infty}^{\infty} f_{XY}(x, y) dy \right\} dx, \quad F_Y(y) = \int_{-\infty}^y \left\{ \int_{-\infty}^{\infty} f_{XY}(x, y) dx \right\} dy$$

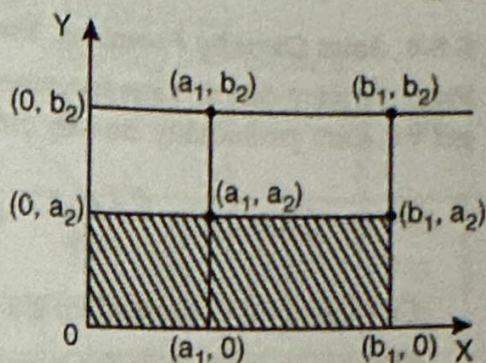


Fig. 5.9a

5.5.4. Joint Density Function, Marginal Density Function

From the joint distribution function $F_{XY}(x, y)$ of two-dimensional continuous random variable, we get the joint probability density function by differentiation as follows :

$$f_{XY}(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y} = \lim_{\delta x \rightarrow 0, \delta y \rightarrow 0} \frac{P(x \leq X \leq x + \delta x, y \leq Y \leq y + \delta y)}{\delta x \delta y} \quad \dots (5-17)$$

Or it may be expressed in the following way also :

"The probability that the point (x, y) will lie in the infinitesimal rectangular region, of area $dx dy$ is given by :

$$P\left(x - \frac{1}{2}dx \leq X \leq x + \frac{1}{2}dx, \quad y - \frac{1}{2}dy \leq Y \leq y + \frac{1}{2}dy\right) = dF_{XY}(x, y) \quad \dots (5-17a)$$

and is denoted by $f_{XY}(x, y) dx dy$, where the function $f_{XY}(x, y)$ is called the joint probability density function of X and Y .

The marginal probability functions of X and Y are given respectively as follows :

$$f_X(x) = \begin{cases} \sum_y p_{XY}(x, y) & \text{(for discrete variables)} \\ \int_{-\infty}^{\infty} f_{XY}(x, y) dy, & \text{(for continuous variables)} \end{cases} \quad \dots (5-17 b)$$

$$\text{and } f_Y(y) = \begin{cases} \sum_x p_{XY}(x, y) & \text{(for discrete variables)} \\ \int_{-\infty}^{\infty} f_{XY}(x, y) dx, & \text{(for continuous variables)} \end{cases} \quad \dots (5-17 c)$$

The marginal density functions of X and Y can be obtained in the following manner also :

$$\text{and } \left. \begin{aligned} f_X(x) &= \frac{dF_X(x)}{dx} = \int_{-\infty}^{\infty} f_{XY}(x, y) dy \\ f_Y(y) &= \frac{dF_Y(y)}{dy} = \int_{-\infty}^{\infty} f_{XY}(x, y) dx \end{aligned} \right\} \quad \dots (5-17 d)$$

Important Remark. If we know the joint p.d.f. (p.m.f.) $f_{XY}(x, y)$ of two random variables X and Y , we can obtain the individual distributions of X and Y in the form of their marginal p.d.f.'s (p.m.f.'s) $f_X(x)$ and $f_Y(y)$ by using (5-17 b) and (5-17 c). However, the converse is not true, i.e., from the marginal distributions of two jointly distributed random variables, we cannot determine the joint distributions of these two random variables.

To verify this, it will suffice to show that two different joint p.m.f.'s (p.d.f.'s) have the same marginal distribution for X and the same marginal distribution for Y . We give below two joint discrete probability distributions which have the same marginal distributions.

Probability Distribution I

X \ Y	0	1	$f_Y(y)$
0	0.28	0.37	0.65
1	0.22	0.13	0.35
$f_X(x)$	0.50	0.50	1.00

Probability Distribution II

X \ Y	0	1	$f_Y(y)$
0	0.35	0.30	0.65
1	0.15	0.20	0.35
$f_X(x)$	0.50	0.50	1.00

As an illustration for continuous r.v.'s, let (X, Y) be continuous r.v. with joint p.d.f. :

$$f_{XY}(x, y) = x + y ; 0 \leq (x, y) \leq 1$$

The marginal p.d.f.'s of X and Y are given by :

... (5-17 e)

$$f_X(x) = \int_0^1 f(x, y) dy = \int_0^1 (x + y) dy = \left[xy + \frac{y^2}{2} \right]_0^1$$

$$\Rightarrow f_X(x) = x + \frac{1}{2} ; 0 \leq x \leq 1$$

$$\text{Similarly, } f_Y(y) = \int_0^1 f(x, y) dx = y + \frac{1}{2} ; 0 \leq y \leq 1$$

... (5-17 f)

Consider another continuous joint p.d.f. :

$$g(x, y) = \left(x + \frac{1}{2} \right) \left(y + \frac{1}{2} \right) ; 0 \leq (x, y) \leq 1$$

... (5-17 g)

Then marginal p.d.f.'s of X and Y are given by :

$$g_1(x) = \int_0^1 g(x, y) dy = \left(x + \frac{1}{2} \right) \int_0^1 \left(y + \frac{1}{2} \right) dy = \left(x + \frac{1}{2} \right) \left[\frac{y^2}{2} + \frac{1}{2}y \right]_0^1$$

$$\Rightarrow g_1(x) = x + \frac{1}{2} ; 0 \leq x \leq 1$$

Similarly,

$$g_2(y) = y + \frac{1}{2} ; 0 \leq y \leq 1$$

... (5-17 h)

(5-17 f) and (5-17 h) imply that the two joint p.d.f.'s in (5-17 e) and (5-17 g) have the same marginal p.d.f.'s (5-17 f) or (5-17 h).

5-5.5. The Conditional Distribution Function and Conditional Probability Density Function

For two-dimensional random variable (X, Y) , the joint distribution function $F_{XY}(x, y)$ for any real numbers x and y is given by :

$$F_{XY}(x, y) = P(X \leq x, Y \leq y)$$

Now let A be the event $(Y \leq y)$ such that the event A is said to occur when Y assumes values upto and inclusive of y . Using conditional probabilities we may now write

$$F_{XY}(x, y) = \int_{-\infty}^x P(A | X = x) dF_X(x) \quad \dots (5-18)$$

The conditional distribution function $F_{Y|X}(y | x)$ denotes the distribution function of Y when X has already assumed the particular value x . Hence,

$$F_{Y|X}(y|x) = P(Y \leq y | X = x) = P(A | X = x)$$

Using this expression, the joint distribution function $F_{XY}(x, y)$ may be expressed in terms of the conditional distribution function as follows :

$$F_{XY}(x, y) = \int_{-\infty}^x F_{Y|X}(y|x) dF_X(x) \quad \dots (5.18 a)$$

Similarly,

$$F_{XY}(x, y) = \int_{-\infty}^y F_{X|Y}(x|y) dF_Y(y) \quad \dots (5.18 b)$$

The conditional probability density function of Y given X for two random variables X and Y which are jointly continuously distributed is defined as follows, for two real numbers x and y :

$$f_{Y|X}(y|x) = \frac{\partial}{\partial y} F(y|x) \quad \dots (5.19)$$

Remarks:

1. If $f_X(x) > 0$, then $f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)}$.

Proof. We have

$$F_{XY}(x, y) = \int_{-\infty}^x f_{Y|X}(y|x) dF_X(x) = \int_{-\infty}^x f_{Y|X}(y|x) f_X(x) dx$$

$$\text{Differentiating w.r. to } x, \quad \frac{\partial}{\partial x} F_{XY}(x, y) = F_{Y|X}(y|x) f_X(x)$$

$$\text{Differentiating w.r. to } y, \text{ we get } \frac{\partial}{\partial y} \left\{ \frac{\partial}{\partial x} F_{XY}(x, y) \right\} = f_{Y|X}(y|x) f_X(x)$$

$$\Rightarrow f_{XY}(x, y) = f_{Y|X}(y|x) f_X(x)$$

$$\therefore f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)}$$

$$2. \text{ If } f_Y(y) > 0, \text{ then } f_{X|Y}(x|y) = \frac{f_{XY}(x, y)}{f_Y(y)}$$

3. In terms of the differentials, we have

$$\begin{aligned} P(x < X \leq x + dx | y < Y \leq y + dy) &= \frac{P(x < X \leq x + dx, y < Y \leq y + dy)}{P(y < Y \leq y + dy)} = \frac{f_{XY}(x, y) dx dy}{f_Y(y) dy} \\ &= f_{X|Y}(x|y) dx \end{aligned}$$

Hence, $f_{X|Y}(x|y)$ may be interpreted as the conditional density function of X on the assumption $Y = y$.

Note. The conditional probability $P(a \leq X \leq b | Y = y)$, as such is meaningless because for a continuous r.v. X , the probability at a point is zero, i.e., $P(Y = y) = 0$, i.e., the event $\{Y = y\}$ is an impossible event. However this difficulty is avoided, by considering ' $Y = y'$ ' as a limiting value :

$$(Y = y) = \lim_{\delta y \rightarrow 0} [y \leq Y \leq y + \delta y].$$

5.5.6. Stochastic Independence

Let us consider two random variables X and Y (of discrete or continuous type) with joint p.d.f. (p.m.f.) $f_{XY}(x, y)$ and marginal p.d.f.'s (p.m.f.'s) $f_X(x)$ and $g_Y(y)$, respectively. Then by the compound probability theorem :

$$f_{XY}(x, y) = f_X(x) g_{Y|X}(y|x)$$

where $g_{Y|X}(y|x)$ is the conditional p.d.f. of Y for given value of $X = x$.

If we assume that $g(y|x)$ does not depend on x , then by the definition of marginal p.d.f.'s, we get for continuous r.v.'s :

$$g(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_{-\infty}^{\infty} f_X(x) g(y|x) dx$$

$$= g(y|x) \int_{-\infty}^{\infty} f_X(x) dx$$

[since $g(y|x)$ does not depend on x]

$$= g(y|x)$$

[$\because f(\cdot)$ is p.d.f. of X]

$$g(y) = g(y|x) \quad \text{and} \quad f_{XY}(x, y) = f_X(x) \cdot g_Y(y)$$

... (*)

Hence

provided $g(y|x)$ does not depend on x . This motivates the following definition of independent random variables.

Independent Random Variables. Two r.v.'s X and Y with joint p.d.f. (p.m.f.) $f_{XY}(x, y)$ and marginal p.d.f.'s (p.m.f.'s) $f_X(x)$ and $g_Y(y)$, respectively, are said to be stochastically independent if and only if

$$f_{XY}(x, y) = f_X(x) \cdot g_Y(y) \quad \dots (5.20)$$

Remarks:

1. In terms of the distribution function, we have the following definition :

Two jointly distributed random variables X and Y are stochastically independent if and only if their joint distribution function $F_{XY}(\cdot, \cdot)$ is the product of their marginal distribution function $F_X(\cdot)$ and $G_Y(\cdot)$, i.e., if for real (x, y)

$$F_{XY}(x, y) = F_X(x) \cdot G_Y(y) \quad \dots (5.20 a)$$

2. The variables which are not stochastically independent are said to be stochastically dependent.

Theorem 5.1. Two random variables X and Y with joint p.d.f. $f(x, y)$ are stochastically independent if and only if $f_{XY}(x, y)$ can be expressed as the product of a non-negative function of x alone and a non-negative function of y alone, i.e., if

$$f_{XY}(x, y) = h_X(x) \cdot k_Y(y), \text{ where } h(\cdot) \geq 0 \text{ and } k(\cdot) \geq 0. \quad \dots (5.20 b)$$

Proof. If X and Y are independent, then by def., $f_{X,Y}(x, y) = f_X(x) \cdot g_Y(y)$,

where $f(x)$ and $g(y)$ are marginal p.d.f.'s of X and Y respectively. Thus condition (5.20b) is satisfied.

Conversely if (5.20b) holds, then we have to prove that X and Y are independent. For continuous random variables X and Y , the marginal p.d.f.'s are given by :

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_{-\infty}^{\infty} h(x) k(y) dy = h(x) \int_{-\infty}^{\infty} k(y) dy = c_1 h(x), \text{ say } \dots (*)$$

$$\text{and } g_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_{-\infty}^{\infty} h(x) k(y) dx = k(y) \int_{-\infty}^{\infty} h(x) dx = c_2 k(y), \text{ say } \dots (**)$$

where c_1 and c_2 are constants independent of x and y . Moreover,

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy &= 1 \quad \Rightarrow \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x) k(y) dx dy = 1 \\ \Rightarrow \left(\int_{-\infty}^{\infty} h(x) dx \right) \left(\int_{-\infty}^{\infty} k(y) dy \right) &= 1 \quad \Rightarrow \quad c_2 c_1 = 1 \quad [\text{From } (**) \text{ and } (*)] \end{aligned}$$

Finally, we get

$$\begin{aligned} f_{XY}(x, y) &= h_X(x) k_Y(y) = c_1 c_2 h_X(x) k_Y(y) \\ &= \{c_1 h_X(x)\} \{c_2 k_Y(y)\} \\ &= f_X(x) g_Y(y) \end{aligned}$$

$\Rightarrow$ X and Y are stochastically independent.

Theorem 5.2. If the random variables X and Y are stochastically independent, then for all possible selections of the corresponding pairs of real numbers (a_1, b_1) , (a_2, b_2) where $a_i \leq b_i$ for all $i = 1, 2$ and where the values $\pm \infty$ are allowed, the events $(a_1 < X \leq b_1)$ and $(a_2 < Y \leq b_2)$ are independent, i.e.,

$$P\{(a_1 < X \leq b_1) \cap (a_2 < Y \leq b_2)\} = P(a_1 < X \leq b_1) \cdot P(a_2 < Y \leq b_2)$$

Proof. Since X and Y are stochastically independent, in the usual notations :

$$f_{X,Y}(x, y) = f_X(x) g_Y(y)$$

In case of continuous r.v.'s, we have

$$\begin{aligned} P[(a_1 < X \leq b_1) \cap (a_2 < Y \leq b_2)] &= \int_{a_1}^{b_1} \int_{a_2}^{b_2} f(x, y) dx dy \\ &= \left(\int_{a_1}^{b_1} f_X(x) dx \right) \left(\int_{a_2}^{b_2} g_Y(y) dy \right) \\ &= P(a_1 < X \leq b_1) P(a_2 < Y \leq b_2), \text{ as desired.} \end{aligned}$$

$\Rightarrow$ The events : $(a_1 < X \leq b_1)$ and $(a_2 < Y \leq b_2)$ are independent.

Remark. In case of discrete r.v.'s theorems 5.2 and 5.3 can be proved on replacing integration by summation over the given range of the variables.

5.5.7. Generalisation to n -Dimensional Random Variable

The concept of two-dimensional random variables and their joint and marginal distributions in § 5.5 to § 5.5.6 can be easily generalised to the case of n -dimensional random variable.

Joint and Marginal Probability Mass Function

Let $(X_1, X_2, \dots, X_n)$ be a discrete n -dimensional r.v., assuming discrete values, in some region, say, R^n of the n -dimensional space. Then the joint p.m.f. of $(X_1, X_2, \dots, X_n)$ is defined as :

$$\begin{aligned} p_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) &= P[X_1 = x_1, X_2 = x_2, \dots, X_n = x_n] \\ &= P\left[\bigcap_{i=1}^n (X_i = x_i)\right] \end{aligned} \quad \dots (5.21)$$

- where,
- (i) $p(x_1, x_2, \dots, x_n) \geq 0, \forall (x_1, x_2, \dots, x_n) \in R^n$, and
- (ii) $\sum_{x_1, x_2, \dots, x_n} p(x_1, x_2, \dots, x_n) = 1$

The marginal *p.m.f.* of any *r.v.*, say, X_i , is obtained on summing $p(x_1, x_2, \dots, x_n)$, over the values of all other variables except X_i . Thus,

$$P_{X_i}(x_i) = \sum_{\substack{(x_1, x_2, \dots, x_n) \\ \text{except } x_i}} p(x_1, x_2, \dots, x_n) \quad \dots (5.21a)$$

In particular, if $p(x_1, x_2, x_3)$ is the joint *p.m.f.* of three *r.v.*'s X_1, X_2 and X_3 , then the marginal *p.m.f.* of, say, X_1 is given by :

$$P_{X_1}(x_1) = \sum_{x_2, x_3} p(x_1, x_2, x_3), \quad \dots (5.21b)$$

and so on.

As, in the case of two random variables, the *r.v.*'s $X_1, X_2, \dots, X_n$ are independent if and only if their joint *p.m.f.* is equal to the product of their marginal *p.m.f.*'s, i.e., iff :

$$p_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = p_{X_1}(x_1) \cdot p_{X_2}(x_2) \dots p_{X_n}(x_n) \quad \dots (5.21c)$$

Joint and Marginal Probability Density Function

Let $(X_1, X_2, \dots, X_n)$ be *n*-dimensional continuous *r.v.* assuming all the values in some region, say, R_1^n of the *n*-dimensional space. Then the joint *p.d.f.* of $(X_1, X_2, \dots, X_n)$ is given by :

$$f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = \lim_{dx_1 \rightarrow 0, dx_2 \rightarrow 0, \dots, dx_n \rightarrow 0} \frac{p\left[\bigcap_{i=1}^n (x_i < X_i < x_i + dx_i)\right]}{dx_1 \cdot dx_2 \cdot \dots \cdot dx_n} \quad \dots (5.21d)$$

where :

- (i) $f(x_1, x_2, \dots, x_n) \geq 0, \forall (x_1, x_2, \dots, x_n) \in R_1^n$, and

(ii) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f(x_1, x_2, \dots, x_n) dx_1 dx_2 \dots dx_n = 1$

The marginal *p.d.f.* of any variable, say, X_i , is obtained on integrating the joint *p.d.f.* over the range of all the variables except X_i . Thus,

$$f_{X_i}(x_i) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f(x_1, x_2, \dots, x_n) dx_1 \dots dx_{i-1} dx_{i+1} \dots dx_n \quad \dots (5.21e)$$